Polynomials and Vieta's formulas

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L3-AL1 Level 3 Senior · Algebra

How the coefficients of a polynomial encode its roots, and how to use remainders, Vieta's formulas and symmetric expressions to answer questions about roots you never find.
By the end you can find remainders and factors quickly with the remainder and factor theorems; evaluate symmetric expressions in the roots of a polynomial without solving it; use the equation itself to compute power sums and to build polynomials with related roots.
Prerequisites: Factorising, substitution and systems (L2-AL1), Sequences, sums and averages (L2-AL2). **Contests:** UKMT SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 150 minutes.

1 Lesson

The equation $x^3 - 5x^2 + 3x - 7 = 0$ has roots a, b and c. Find $a^2 + b^2 + c^2$.

One could try to solve the cubic. It has no rational roots (the only candidates are ± 1 and ± 7, and none of them works), one real root near 4.68 and two complex roots. Squaring those is impractical. Instead, note that the cubic factorises as $(x-a)(x-b)(x-c)$, and expanding that and comparing with the given cubic tells us $a+b+c = 5$ and $ab+bc+ca = 3$. Then

$$a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca) = 25 - 6 = 19.$$

We never found a root. This unit is about extracting information on the roots from the coefficients alone, and contest problems of this type are set so that this is all you need.

1.1 Division, remainders and the factor theorem

We start with polynomial division. If P is a polynomial and D is a non-zero polynomial, there are unique polynomials Q (the quotient) and R (the remainder) with

$$P(x) = D(x)\,Q(x) + R(x), \qquad \deg R < \deg D.$$

This is long division, just as with integers, and "the degree of the remainder is less than the degree of the divisor" plays the role of "the remainder is smaller than the divisor".

Remainder theorem.
The remainder when $P(x)$ is divided by $x - k$ is $P(k)$.
Factor theorem. $x - k$ is a factor of $P(x)$ if and only if $P(k) = 0$.

The proof is two lines. Dividing by $x - k$, which has degree 1, leaves a remainder of degree 0, that is a constant r. So $P(x) = (x-k)Q(x) + r$ for every x. Put $x = k$: the first term vanishes and $P(k) = r$. The factor theorem is the case $r = 0$.

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Three habits follow. First, dividing by $x+3$ means $k=-3$, so the remainder is $P(-3)$. Second, when you divide by a quadratic the remainder has the form $px+q$, which has two unknowns, so you need two facts to pin it down. If the quadratic has distinct roots α and β, those facts are $P(\alpha)$ and $P(\beta)$, because the $D(x)Q(x)$ term vanishes at both. Third, if you know the values of a polynomial at several points, subtract something simple that takes the same values. The difference has known roots, so the factor theorem determines it once you know its degree and leading coefficient.

1.2 Vieta's formulas

Suppose $P(x)=ax^3+bx^2+cx+d$ with $a\neq 0$ has roots r, s, t, where repeated roots are listed as often as they occur and complex roots are allowed. By the factor theorem, used three times, $P(x)=a(x-r)(x-s)(x-t)$. Expand the right-hand side:

$$a(x-r)(x-s)(x-t)=a\big(x^3-(r+s+t)x^2+(rs+st+tr)x-rst\big).$$

Now compare coefficients with ax^3+bx^2+cx+d.

Vieta's formulas.
If ax^3+bx^2+cx+d has roots r,s,t, then

$$r+s+t=-\frac{b}{a},\qquad rs+st+tr=\frac{c}{a},\qquad rst=-\frac{d}{a}.$$

For a quadratic ax^2+bx+c with roots r,s we get $r+s=-\frac{b}{a}$ and $rs=\frac{c}{a}$. In degree n, the sum of the products of the roots taken k at a time equals $(-1)^k$ times the coefficient of x^{n-k}, divided by the leading coefficient.

The signs alternate, starting with a minus, and you always divide by the leading coefficient. I will write $e_1=r+s+t$, $e_2=rs+st+tr$ and $e_3=rst$. These are the *elementary symmetric polynomials* of the roots. (That a polynomial of degree n has exactly n complex roots, counted with multiplicity, is the Fundamental Theorem of Algebra. You may quote it.)

1.3 Symmetric expressions

An expression in r,s,t is *symmetric* if swapping any two of the letters leaves it unchanged, like $r^2+s^2+t^2$ or $\frac{1}{r}+\frac{1}{s}+\frac{1}{t}$. A classical theorem says that every symmetric polynomial in the roots can be written as a polynomial in e_1,e_2,e_3, so every such expression can be computed from the coefficients alone. In practice you need only the following identities, each of which you can check by expanding:

- $r^2+s^2+t^2=e_1^2-2e_2$;
- $\frac{1}{r}+\frac{1}{s}+\frac{1}{t}=\frac{e_2}{e_3}$;
- $r^3+s^3+t^3=e_1^3-3e_1e_2+3e_3$;
- $(k-r)(k-s)(k-t)=\frac{P(k)}{a}$ for any number k.

The last line is often overlooked, and it is frequently the fastest. It is the factorisation $P(x)=a(x-r)(x-s)(x-t)$ evaluated at $x=k$.

Higher power sums $p_n=r^n+s^n+t^n$ come from the equation itself. For a monic cubic x^3+bx^2+cx+d, each root satisfies $r^3+br^2+cr+d=0$. Multiply by r^{n-3} and add the three versions of

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this equation:

$$p_n + b\,p_{n-1} + c\,p_{n-2} + d\,p_{n-3} = 0 \qquad (n \geq 3),$$

where $p_0 = 3$. This is a form of Newton's identities, and it computes power sums by recurrence.

1.4 New roots from old

Sometimes you want a polynomial whose roots are related to the old ones. If P has roots r_i, then $P(x-2)$ has roots $r_i + 2$, $P(x/3)$ has roots $3r_i$, and reversing the coefficients of P gives a polynomial with roots $\frac{1}{r_i}$ (provided no root is 0). Once you have the new polynomial, Vieta applies to it as usual.

Example 1.
The roots of $x^2 - 7x + 3 = 0$ are p and q. Find $\frac{p}{q} + \frac{q}{p}$.

Solution.
The quadratic formula gives $\frac{7 \pm \sqrt{37}}{2}$, but dividing surds by surds is slow and error-prone. Note instead that the target does not change if we swap p and q, so it must be expressible through $p + q = 7$ and $pq = 3$.
Combining the fractions, $\frac{p}{q} + \frac{q}{p} = \frac{p^2+q^2}{pq} = \frac{(p+q)^2-2pq}{pq} = \frac{49-6}{3} = \frac{43}{3}$.
When the target is symmetric, rewrite it in terms of the sum and product before doing anything else.

Example 2.
When $P(x)$ is divided by $x^2 - 4$ the remainder is $3x + 1$, and when $P(x)$ is divided by $x - 3$ the remainder is 15. Find the remainder when $P(x)$ is divided by $(x^2 - 4)(x - 3)$.

Solution.
The divisor is a cubic, so the remainder $R(x)$ has degree at most 2. The direct approach is to write $R(x) = ux^2 + vx + w$ and find three equations. We know $P(2) = 7$, $P(-2) = -5$ and $P(3) = 15$, and R agrees with P at the roots $2, -2, 3$ of the divisor, so this system can be solved. This works, but a shorter route uses the given remainder directly.
Write $P(x) = (x^2 - 4)(x - 3)Q(x) + R(x)$. The first term is a multiple of $x^2 - 4$, so P and R leave the same remainder on division by $x^2 - 4$, namely $3x + 1$. Since R has degree at most 2, its quotient on division by $x^2 - 4$ is a constant, so $R(x) = c(x^2 - 4) + 3x + 1$ for some number c. Now there is only one unknown.
Putting $x = 3$ gives $15 = P(3) = R(3) = 5c + 10$, so $c = 1$ and the remainder is $x^2 + 3x - 3$. As a check, at $x = 2$ this gives $4 + 6 - 3 = 7$, which matches $3 \cdot 2 + 1$.

Example 3.
The roots of $x^3 - 4x^2 + x + 5 = 0$ are a, b and c. Find $(a + b)(b + c)(c + a)$.

Solution.
One could expand. The product is $a^2b + ab^2 + b^2c + bc^2 + c^2a + ca^2 + 2abc$, and with some effort that equals $e_1e_2 - e_3$. Here $e_1 = 4$, $e_2 = 1$ and $e_3 = -5$, so the answer is $4 + 5 = 9$. This needs an identity that is easy to forget in an exam.
It is quicker to note that each bracket is the sum of all the roots with one left out. Since $a + b + c = 4$, we have $a + b = 4 - c$, and similarly for the others. So the product is $(4 - a)(4 - b)(4 - c)$, which by the last identity above is $P(4) = 64 - 64 + 4 + 5 = 9$.

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In general, a bracket containing all but one root should be rewritten as e_1 minus that root, and then a product of such brackets is a value of P.

Example 4.
Let a, b, c be the roots of $x^3 - x - 1 = 0$. Find $a^5 + b^5 + c^5$.

Solution.
Expanding $(a+b+c)^5$ and subtracting the unwanted terms is impractical, and the roots themselves are no help: one is about 1.3247 and the other two are complex. Instead we use the equation to reduce powers (Newton's identities). Each root satisfies $r^3 = r + 1$, and multiplying by r^{n-3} gives $r^n = r^{n-2} + r^{n-3}$. Adding this over the three roots gives $p_n = p_{n-2} + p_{n-3}$.
We need starting values. Clearly $p_0 = 3$, and $p_1 = e_1 = 0$ because there is no x^2 term. Also $e_2 = -1$, so $p_2 = e_1^2 - 2e_2 = 2$. Then $p_3 = p_1 + p_0 = 3$, $p_4 = p_2 + p_1 = 2$ and $p_5 = p_3 + p_2 = 5$.
As a check, $1.3247^5 \approx 4.08$, so the complex pair should contribute about 0.92, which is real, as the contribution of a conjugate pair must be. The answer is 5.

Common traps.

- Sign errors. For $ax^3 + bx^2 + cx + d$ the sum of the roots is $-\frac{b}{a}$, and the product is $-\frac{d}{a}$, whereas for a quadratic the product is $+\frac{c}{a}$. If in doubt, expand $(x-r)(x-s)(x-t)$ again.
- Forgetting the leading coefficient. For $2x^2 - 6x + 1$ the product of the roots is $\frac{1}{2}$, not 1.
- Missing terms. In $x^3 - 7x + 6$ the coefficient of x^2 is 0, so the roots add up to 0.
- Vieta counts every root, complex and repeated ones included. If the question asks for the sum of the *real* roots or of the *distinct* roots, find out which roots those are first.
- A negative value for $r^2 + s^2 + t^2$ is not an error. It means some roots are not real.
- The remainder on division by $x + k$ is $P(-k)$, not $P(k)$.

1.5 How to spot this in a contest

The usual signals are these. A question names the roots of a polynomial but the roots are ugly, or the polynomial has no obvious factor. The thing you are asked for is symmetric in the roots. You are told remainders, or the values $P(1), P(2), \ldots$ of a polynomial of known degree, in which case look at $P(x)$ minus something simple. A product such as $(2-r)(2-s)(2-t)$ appears, which is a value of P. In every case, do not solve; translate into e_1, e_2, e_3 or into values of P, and compute.

On the SMC these questions are quick once recognised, so read the coefficients before trying anything else. On the AMC 12 and the AIME the same tools appear with an extra step: a substitution to tidy the coefficients, a root that can be guessed, or two polynomials whose roots are related. If a calculation is getting long, ask whether you have missed a symmetric shortcut.

2 Practice problems

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The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ The roots of $2x^2 - 6x + 1 = 0$ are r and s. Find $r^2 + s^2$.

Problem 2 ⋆⋆ Find the remainder when $x^{2026} + 3x^{101} - 5$ is divided by $x^2 - 1$.

Problem 3 ⋆⋆ The equation $x^3 - 6x^2 + kx + 10 = 0$ has three real roots that form an arithmetic progression. Find k.

Problem 4 ⋆⋆⋆ A polynomial $P(x)$ of degree 4 with leading coefficient 1 satisfies $P(1) = 2$, $P(2) = 5$, $P(3) = 10$ and $P(4) = 17$. Find $P(5) + P(0)$.

Problem 5 ⋆⋆⋆ Let a, b and c be the roots of $x^3 - 2x^2 + 3x - 4 = 0$. Find $(a^2+1)(b^2+1)(c^2+1)$.

Problem 6 ⋆⋆⋆⋆ Let a, b and c be the roots of $x^3 - 4x + 1 = 0$. Find

$$\frac{a^2}{a+2} + \frac{b^2}{b+2} + \frac{c^2}{c+2}.$$

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT SMC 2015 Question 2** ⋆
   Clear the fraction, then read the sum straight off the quadratic
2. **AMC 10A 2003 Problem 18** ⋆
   Clear the fraction to get a quadratic, then read off what you need
3. **AHSME 1999 Problem 17** ⋆⋆
   The remainder is linear, and you know its value at two points
4. **UKMT SMC 2024 Question 11** ⋆⋆
   Pair the outer and the inner factors so that both brackets share a common quadratic
5. **UKMT SMC 2023 Question 18** ⋆⋆
   Treat the equation as a cubic in $\sin X$ *and look for a factor*
6. **AMC 12A 2022 Problem 15** ⋆⋆
   The new volume is a product of brackets of the form root plus a constant
7. **AMC 12B 2004 Problem 17** ⋆⋆⋆
   A sum of logarithms is the logarithm of a product
8. **UKMT SMC 2022 Question 19** ⋆⋆⋆
   Subtract the two equations and factorise, then count the roots of each resulting quadratic with care
9. **AIME 1996 Problem 5** ⋆⋆⋆
   Each new root is the sum of all the old roots minus one of them

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10. **AIME II 2008 Problem 7** ⋆⋆⋆
    Rewrite each bracket using the sum of the roots, then use a cube identity

11. **AIME I 2014 Problem 9** ⋆⋆⋆⋆
    A substitution tidies the coefficients, and then one root can be found by hand

12. **AIME II 2014 Problem 5** ⋆⋆⋆⋆
    Both cubics have no x^2 term, which tells you each one's third root

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4 Hints

1. Write r^2+s^2 using $r+s$ and rs, and remember to divide by the leading coefficient.
2. The remainder has the form $px+q$. Which two values of x make x^2-1 vanish?
3. Call the roots $m-d$, m and $m+d$. What does the sum of the roots tell you about m?
4. The given values look like n^2+1. What can you say about $P(x)-(x^2+1)$?
5. Factorise a^2+1 using i, then remember that $(k-a)(k-b)(k-c)=P(k)$ for a monic cubic P.
6. First divide: $\frac{a^2}{a+2}=a-2+\frac{4}{a+2}$. Then find a cubic whose roots are $a+2$, $b+2$ and $c+2$.

5 Solutions

Problem 1

The roots of $2x^2-6x+1=0$ are r and s. Find r^2+s^2.

By Vieta's formulas, $r+s=\frac{6}{2}=3$ and $rs=\frac{1}{2}$. Then $r^2+s^2=(r+s)^2-2rs=9-1=8$.

Answer: 8

Problem 2

Find the remainder when $x^{2026}+3x^{101}-5$ is divided by x^2-1.

Write $x^{2026}+3x^{101}-5=(x^2-1)Q(x)+px+q$. Putting $x=1$ gives $1+3-5=p+q$, so $p+q=-1$. Putting $x=-1$, and using that 2026 is even and 101 is odd, gives $1-3-5=-p+q$, so $q-p=-7$. Adding the two equations gives $2q=-8$, so $q=-4$ and $p=3$. The remainder is $3x-4$.

Alternatively, when working modulo x^2-1 we may replace x^2 by 1, so $x^{2026}=(x^2)^{1013}$ becomes 1 and $x^{101}=x(x^2)^{50}$ becomes x, leaving $1+3x-5$.

Answer: $3x-4$

Problem 3

The equation $x^3-6x^2+kx+10=0$ has three real roots that form an arithmetic progression. Find k.

Let the roots be $m-d$, m and $m+d$. Their sum is $3m$, and by Vieta it equals 6, so $m=2$ and 2 is a root. Substituting $x=2$ gives $8-24+2k+10=0$, so $k=3$.

We should check that this value really works. Indeed $x^3-6x^2+3x+10=(x-2)(x^2-4x-5)=(x-2)(x+1)(x-5)$, whose roots -1, 2, 5 are real and form an arithmetic progression. So $k=3$.

Answer: 3

Problem 4

A polynomial $P(x)$ of degree 4 with leading coefficient 1 satisfies $P(1)=2$, $P(2)=5$, $P(3)=10$ and $P(4)=17$. Find $P(5)+P(0)$.

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The given values are 1^2+1, 2^2+1, 3^2+1 and 4^2+1. So the polynomial $Q(x) = P(x) - (x^2+1)$ vanishes at $x = 1, 2, 3, 4$. Subtracting a quadratic does not change the x^4 term, so Q has degree 4 and leading coefficient 1. By the factor theorem, $Q(x) = (x-1)(x-2)(x-3)(x-4)$.

Hence $P(5) = 26 + 4 \cdot 3 \cdot 2 \cdot 1 = 50$ and $P(0) = 1 + (-1)(-2)(-3)(-4) = 25$, so $P(5) + P(0) = 75$. Note that the tempting guess $P(x) = x^2 + 1$ is not allowed, because P must have degree 4.

Answer: 75

Problem 5

Let a, b and c be the roots of $x^3 - 2x^2 + 3x - 4 = 0$. Find $(a^2+1)(b^2+1)(c^2+1)$.

Write $P(x) = x^3 - 2x^2 + 3x - 4 = (x-a)(x-b)(x-c)$. Since $a^2 + 1 = (i-a)(-i-a)$, the product we want is $(i-a)(i-b)(i-c) \cdot (-i-a)(-i-b)(-i-c) = P(i)\,P(-i)$. Now $P(i) = -i + 2 + 3i - 4 = -2 + 2i$ and $P(-i) = i + 2 - 3i - 4 = -2 - 2i$, so the answer is $(-2+2i)(-2-2i) = 4 + 4 = 8$.

If you prefer to avoid complex numbers, expand instead: the product equals $a^2b^2c^2 + (a^2b^2 + b^2c^2 + c^2a^2) + (a^2 + b^2 + c^2) + 1$, which is $e_3^2 + (e_2^2 - 2e_1e_3) + (e_1^2 - 2e_2) + 1$. With $e_1 = 2$, $e_2 = 3$ and $e_3 = 4$ this is $16 - 7 - 2 + 1 = 8$.

Answer: 8

Problem 6

Let a, b and c be the roots of $x^3 - 4x + 1 = 0$. Find

$$\frac{a^2}{a+2} + \frac{b^2}{b+2} + \frac{c^2}{c+2}.$$

No root equals -2, since $P(-2) = -8 + 8 + 1 = 1 \neq 0$, so every fraction makes sense. Polynomial division gives $a^2 = (a-2)(a+2) + 4$, so $\frac{a^2}{a+2} = a - 2 + \frac{4}{a+2}$. Adding the three versions, the required sum is $(a+b+c) - 6 + 4S$, where $S = \frac{1}{a+2} + \frac{1}{b+2} + \frac{1}{c+2}$. By Vieta $a + b + c = 0$, because there is no x^2 term.

To find S, note that $y = x + 2$ turns the roots a, b, c into $a+2, b+2, c+2$. These are the roots of $P(y-2) = (y-2)^3 - 4(y-2) + 1 = y^3 - 6y^2 + 8y + 1$. For this cubic, the sum of the reciprocals of the roots is $\frac{e_2}{e_3} = \frac{8}{-1} = -8$. So $S = -8$ and the required sum is $0 - 6 + 4(-8) = -38$.

Answer: -38